

Statistical_Modeling_Report

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Objective: Build a Lie Detector Model

Purpose:

The goal of this project is to predict which participants will successfully complete the program based on their entrance assessment scores:

- 1 = Successfully completed the program through its duration (completer)
- 0 = Unsuccessfully completed the program prior to expiration (non-completer)

The central question driving this project is:

Can I build a model that beats the baseline accuracy of 59% derived from the Bernoulli distribution of the outcome variable?

Tasks:

- **Cleaning Data:** Remove or correct inaccurate records from the dataset.
- **Formatting Data:** Standardize data formats to ensure consistency.

Columns and Their Acronyms:

- **PWS** — Permanency / Navigating Welfare System
- **DL** — Daily Living
- **SC** — Self Care
- **RC** — Relationships & Communication
- **HMM** — Housing & Money Management
- **WSL** — Work & Study Life
- **CEP** — Career & Education Planning
- **LF** — Looking Forward

```
# Load and format the data
my_data = read.csv("/Users/oakmoreroadinc./Desktop/Data Science /Portfolio Data Science /R_script.R/data.csv")
head(my_data)
```

```
## Youth.ID Gender PWS DL SC RC HMM WSL CEP LF Days_in_Program
## 1 1 M 3.80 4.47 4.41 4.33 3.96 4.10 5.00 5.00 264
## 2 2 M 4.00 4.41 4.53 4.78 3.70 4.40 4.10 4.00 467
## 3 3 M 4.30 4.65 4.29 4.78 3.78 4.60 4.11 4.50 49
## 4 4 M 3.55 2.24 2.71 3.83 2.09 2.70 2.11 3.25 908
## 5 5 F 4.90 4.71 4.76 4.83 3.35 4.65 4.78 5.00 241
## 6 6 M 4.50 4.59 4.65 4.83 4.09 4.35 3.89 4.62 495
## Category Result.Score
## 1 Good 1
```

```
## 2      Good      1
## 3      Bad       0
## 4      Good      1
## 5      Bad       0
## 6      Bad       0
```

```
# Find duplicates: How many duplicated rows exist in the df?
duplicate_rows <- sum(duplicated(my_data))
cat("There are", duplicate_rows, "duplicate rows.\n")
```

```
## There are 0 duplicate rows.
```

```
# Finding missing values for all columns
total_missing_values <- sum(is.na(my_data))
cat("There are", total_missing_values, "missing values in total.\n")
```

```
## There are 0 missing values in total.
```

```
colnames(my_data)
```

```
## [1] "Youth.ID"      "Gender"        "PWS"           "DL"
## [5] "SC"            "RC"            "HMM"           "WSL"
## [9] "CEP"           "LF"            "Days_in_Program" "Category"
## [13] "Result.Score"
```

Checking Data variation

```
assess_cols <- c("SC", "CEP", "RC", "LF", "PWS", "HMM", "DL", "WSL")

summary_stats <- data.frame(
  mean = sapply(my_data[, assess_cols], mean, na.rm = TRUE),
  sd    = sapply(my_data[, assess_cols], sd,    na.rm = TRUE)
)

ro((summary_stats), 2)
```

```
##      mean    sd
## SC  4.40 0.62
## CEP 3.99 0.82
## RC   4.55 0.48
## LF   4.52 0.56
## PWS 4.12 0.74
## HMM 3.85 0.72
## DL   4.39 0.68
## WSL 4.24 0.66
```

```
# Add overall ratio of max/min SD
ratio <- max(summary_stats$sd, na.rm=TRUE) / min(summary_stats$sd, na.rm=TRUE)
cat("Max/Min SD ratio:", round(ratio, 2), "\n")
```

```
## Max/Min SD ratio: 1.71
```

I first, pick the columns that hold the assessment scores (SC, CEP, RC, LF, PWS, HMM, DL, WSL). Next, I calculate the **average (mean)** and **spread (standard deviation)** for each assessment. Then, I found the **largest SD** and the **smallest SD**, and divided them to get a ratio. This ratio tells me how uneven the spreads are. A ratio close to 1 means all assessments vary about the same. A high ratio (like 1.7) means one assessment varies much more than another, so you should standardize before transforming to PCA.

Leak-free LOOCV for plain logistic regression (baseline)

```
#makes the results reproducible
set.seed(1)
#list of featute column names
assess_cols <- c("PWS", "DL", "SC", "RC", "HMM", "WSL", "CEP", "LF")

#stores the target variable
y <- my_data$Result.Score

#n holds the number of rows in your dataset.needed later for looping
n <- nrow(my_data)

#creates an empty numeric vector to store predicted probabilities for each row
#as we loop through it fills with predictions
pred_prob_glm <- numeric(n)

#start of loop
#training and test(only one row)separation
for (i in 1:n) {
  tr <- setdiff(1:n, i); te <- i

  # TRAIN data frame (no scaling to match your original glm)
  tr_df <- data.frame(Result.Score = y[tr],
                      my_data[tr, assess_cols, drop = FALSE])

  # Fit logistic model on TRAIN only (suppress numerical separation warnings)
  m_i <- suppressWarnings(glm(Result.Score ~ .,
                             data    = tr_df,
                             family   = binomial,
                             control  = glm.control(maxit = 1000, epsilon = 1e-8))
  )

  # TEST row with same columns
  te_df <- my_data[te, assess_cols, drop = FALSE]

  # Predict probability for held-out row
  pred_prob_glm[i] <- as.numeric(predict(m_i, newdata = te_df, type = "response"))
}
```

```

# Metrics
#Converts probabilities into class predictions (1 or 0).
pred_cls_glm <- as.integer(pred_prob_glm >= 0.5)

#Compares predictions with actuals and calculates the percentage of correct predictions.
loocv_acc_glm <- mean(pred_cls_glm == y)

#error rate
loocv_err_glm <- 1 - loocv_acc_glm

#calculates a baseline accuracy.If you just guesses the majority class
baseline_acc <- max(mean(y == 1), mean(y == 0))

cat("\n--- LOOCV (logistic baseline, no scaling) ---\n")

##
## --- LOOCV (logistic baseline, no scaling) ---

cat("Accuracy:", round(loocv_acc_glm, 3), " Error:", round(loocv_err_glm, 3), "\n")

## Accuracy: 0.545 Error: 0.455

cat("Baseline:", round(baseline_acc, 3), "\n")

## Baseline: 0.591

#creates confusion matrix
print(table(Predicted = pred_cls_glm, Actual = y))

##
##      Actual
## Predicted 0 1
##          0 5 6
##          1 4 7

```

Small-Sample Logistic Model without Leakage

Why this strategy

- With only 22 rows, a traditional train/test split would waste data and give unstable results.
- A **logistic regression** with a few predictors is a strong, interpretable baseline for binary classification.
- **Leave-One-Out Cross-Validation (LOOCV)** trains on $n - 1$ rows and tests on the single held-out row, giving a fairer estimate of out-of-sample performance.

Why this does not leak data

- The model is fit once on the full data only to inspect coefficients and get a descriptive feel.
- Performance is then evaluated through LOOCV, where the held-out observation is never used during model fitting or threshold selection.

What the metrics mean

- **LOOCV accuracy (54.5%)** → The true estimate of how well the model performs on unseen data.
- **Baseline accuracy (59.1%)** → Always predicting the majority class would have done slightly better than this logistic model.

Takeaway

The plain logistic regression **does not generalize well** on this small dataset. It performed worse than a simple baseline in LOOCV.

PCA : GAM using the same pipeline + leak-free LOOCV

```
# =====  
# A) IN-SAMPLE (DIAGNOSTIC):  
# =====  
set.seed(1)  
  
assess_cols <- c("PWS", "DL", "SC", "RC", "HMM", "WSL", "CEP", "LF")  
y <- my_data$Result.Score  
n <- nrow(my_data)  
  
# standardize numeric assessment variables  
my_std <- scale(my_data[, assess_cols])  
  
# PCA on standardized data  
pca_all <- prcomp(as.matrix(my_std), center = FALSE, scale. = FALSE)  
  
# Variance explained  
pv_all <- summary(pca_all)$importance["Proportion of Variance", ]  
  
# Keep components explaining 90% variance  
k_full <- which(cumsum(pv_all) >= 0.90)[1]  
  
# Extract first k_full PCs  
PC_full <- as.data.frame(pca_all$x[, 1:k_full, drop = FALSE])  
colnames(PC_full) <- paste0("PC", 1:k_full)  
  
# Ensure Gender is treated as a categorical factor  
my_data$Gender <- as.factor(my_data$Gender)  
  
# Combine PCs + Gender + Target  
df_full <- cbind(Result.Score = y, PC_full, Gender = my_data$Gender)  
  
# === Build GAM model dynamically ===  
pc_cols <- grep("^PC\\d+$", names(df_full), value = TRUE)  
  
# Drop or keep whichever PCs you want  
drop <- c("PC2", "PC3") # e.g. c("PC4", "PC5") to remove specific PCs  
kept <- setdiff(pc_cols, drop)
```

```

# Choose which PCs to spline
spline <- c("PC1") # flexible, your choice
linear <- setdiff(kept, spline)

# Add Gender to the formula
fit_formula <- as.formula(
  paste(
    "Result.Score ~",
    paste(c(paste0("s(", spline, ")"), linear, "Gender"), collapse = " + ")
  )
)

# === Fit GAM model ===

gam_model <- mgcv::gam(fit_formula, data = df_full, family = binomial())

# === Diagnostics ===
cat("\n--- IN-SAMPLE GAM (PCA + Gender) ---\n")

```

```

##
## --- IN-SAMPLE GAM (PCA + Gender) ---

```

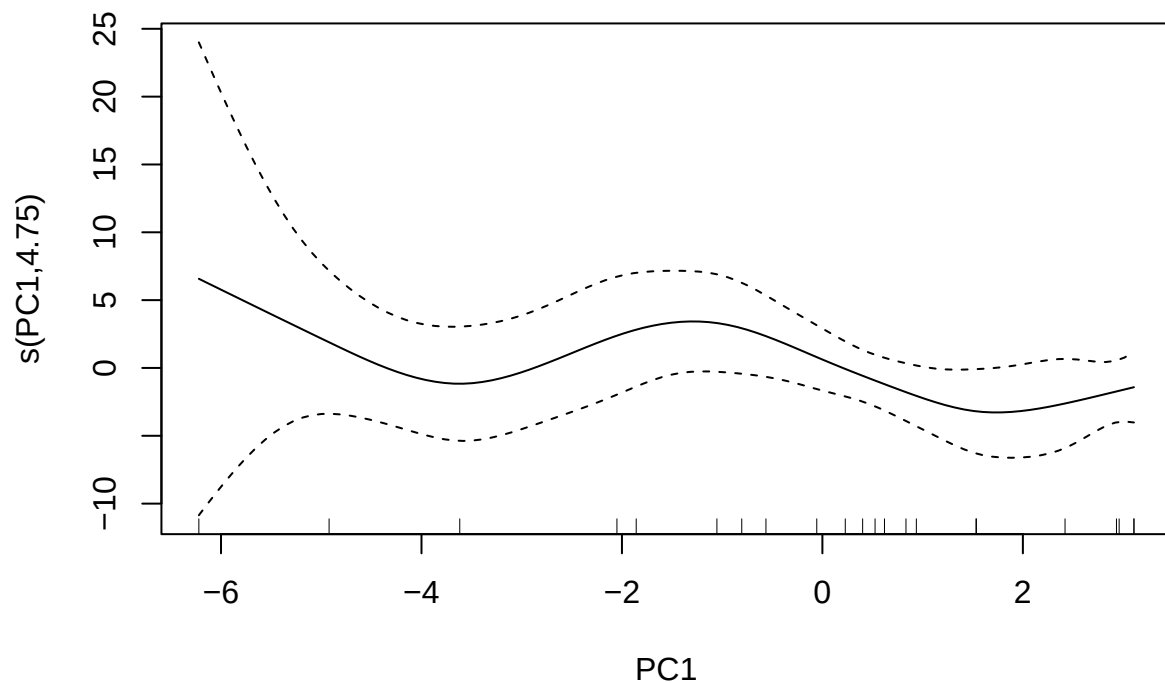
```
summary(gam_model)
```

```

##
## Family: binomial
## Link function: logit
##
## Formula:
## Result.Score ~ s(PC1) + Gender
##
## Parametric coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   2.479      1.244   1.993  0.0463 *
## GenderM       -2.346      1.420  -1.653  0.0984 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##             edf Ref.df Chi.sq p-value
## s(PC1) 4.751  5.756  6.693  0.325
##
## R-sq.(adj) = 0.449  Deviance explained = 49.3%
## UBRE = 0.29993  Scale est. = 1          n = 22

```

```
plot.gam(gam_model)
```



What I Tried

- **Predictors Evaluated:** PC1, PC2, and PC3
- **Smoothing:** PC1 modeled with a spline $s(PC1)$
- PC2 and PC3 were also tested with smooth terms.
- **Categorical Variable:** Gender included as a factor predictor.

Results

- **PC1:** Showed a moderately smooth trend ($edf = 4.751$), but the nonlinearity was **not strongly significant** ($p = 0.325$).
→ Kept PC1, since it captured interpretable structure in the data.
- A curved shape indicates a nonlinear effect — certain ranges of PC1 increase or decrease the probability more sharply.
- **PC2 & PC3:** No significant contribution.
- **Gender:** Borderline effect ($p = 0.0984$), suggesting a potential weak influence.

- **Overall Fit:**
 - Deviance explained: ~**49.3%**
 - Adjusted R²: **0.449**
-

Why I Did This Diagnostic First

Instead of immediately doing train/test splits or LOOCV:

I wanted to **identify which variables actually matter** before committing to a full leak-free validation pipeline. This helps interpret **variable behavior** — significance, shape, and contribution — *in-sample*, before worrying about generalization. It also **reduces model complexity early**, avoiding unnecessary validation of uninformative predictors.

```
glm_check <- glm(Result.Score ~ PC1 + Gender, data = df_full, family = binomial)
summary(glm_check)
```

```
##
## Call:
## glm(formula = Result.Score ~ PC1 + Gender, family = binomial,
##      data = df_full)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.5017     1.0137   1.481  0.1385
## PC1          -0.4976     0.2793  -1.782  0.0748 .
## GenderM       -1.3930     1.1694  -1.191  0.2335
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 29.767  on 21  degrees of freedom
## Residual deviance: 24.984  on 19  degrees of freedom
## AIC: 30.984
##
## Number of Fisher Scoring iterations: 4
```

Sanity Check: Comparing GAM vs. GLM

Why I Ran a Sanity Check

Before finalizing the nonlinear GAM model, I ran a simpler **logistic regression (GLM)** as a sanity check. The purpose was to test whether the additional spline flexibility on PC1 actually improved model performance or interpretability.

- **GAM model:** `Result.Score ~ s(PC1) + Gender`
 - Allows for a nonlinear relationship between PC1 and the outcome.

- **GLM model:** `Result.Score ~ PC1 + Gender`
→ Assumes a straightforward linear relationship.

Running both lets me confirm whether the “curve” that GAM fits provides real predictive benefit, or just adds unnecessary complexity.

Results Summary

Model	Key Findings	Notes
GAM	Smooth term on PC1 not significant (p = 0.325). Gender borderline (p = 0.098).	Explained ~49% deviance but risked overfitting due to small n (22).
GLM	PC1 borderline significant (p = 0.075), Gender weaker (p = 0.23).	Clean convergence, stable estimates, AIC = 31.

Interpretation

- The GAM’s smooth curve on PC1 did **not add meaningful predictive power**.
 - With a **small dataset (n = 22)**, GAM’s spline term introduced unnecessary flexibility, risking overfitting.
 - The GLM captured essentially the **same relationship** — simpler, more stable, and easier to interpret.
-

Conclusion

For this dataset: - The **GLM (linear model)** makes more sense to use.
- It provides nearly identical performance with **less risk of overfitting**, clearer interpretation, and cleaner inference on key predictors (PC1, Gender).

Final model chosen:

`Result.Score ~ PC1 + Gender`

Next Steps

To confirm which model generalizes better beyond the training data, the next step is to use **Leave-One-Out Cross-Validation (LOOCV)** on both models:

1. **GLM Model** — Test the simpler linear model using:
`Result.Score ~ PC1 + Gender`
→ Checks whether PC1 alone (with Gender) provides enough predictive power out-of-sample.

2. **GAM Model** — Test the more flexible nonlinear model using:

Result.Score ~ s(PC1) + PC2 + PC3 + Gender

→ Evaluates whether adding curvature (spline on PC1) and extra components (PC2, PC3) improves accuracy and reduces classification errors.

Comparing LOOCV performance for both models will reveal: - Whether the extra flexibility and complexity of the GAM are justified, and

- If the simpler GLM remains competitive in real-world prediction scenarios.

LOOCV (LEAK-FREE): PC1 + Gender only with GLM model

```
#creates an empty numeric vector of length n
#This will store the predicted probabilities for each left-out observation during LOOCV.
pred_prob <- numeric(n)

# Starts a loop. Use all rows except i for training (tr).
# Use row i for testing (te)
for (i in 1:n) {
  #tr = all indices except i (training set)
  #te = the single index i (test observation)
  tr <- setdiff(1:n, i); te <- i

  # ---- TRAIN: numeric matrix + scale (DO NOT use/overwrite my_std)
  # Extracts only the numeric predictor columns
  # PCA Requires Numeric
  Xtr_raw <- as.matrix(data.frame(lapply(my_data[tr, assess_cols], as.numeric)))

  # scale() standardizes features to zero mean and unit variance.
  # mu and sg store the training set's mean and sd for each column.
  # The guard line handles zero-variance columns to avoid division by zero.
  # This ensures the test data is standardized using training statistics.
  Xtr_sc <- scale(Xtr_raw)
  mu <- attr(Xtr_sc, "scaled:center")
  sg <- attr(Xtr_sc, "scaled:scale"); sg[sg == 0 | is.na(sg)] <- 1

  # ---- PCA on TRAIN standardized data
  pca_tr <- prcomp(Xtr_sc, center = FALSE, scale. = FALSE)

  #Extracts how much variance each principal component explains.
  #kcv = smallest number of PCs explaining at least 90% of total variance.
  #Dynamically selects dimensionality per fold, preventing overfitting.
  pv <- summary(pca_tr)$importance["Proportion of Variance", ]
  kcv <- which(cumsum(pv) >= 0.90)[1]

  #Extracts the new coordinate values of training data (principal component scores).
  #Renames them PC1, PC2, ..., for clarity.
  #These are the numeric predictors that will go into the model.
  trPC <- as.data.frame(pca_tr$x[, 1, drop = FALSE])
  colnames(trPC) <- "PC1"

  # ---- Add Gender to TRAIN
  # Build training dataframe for model fitting.
```

```

# Includes:
# - Result.Score: binary target (0/1)
# - PC columns from PCA
# - Gender: categorical factor (M/F)
# This will be used in the GAM/GLM model.
trDF <- data.frame(
  Result.Score = y[tr],
  PC1 = trPC$PC1,
  Gender = as.factor(my_data$Gender[tr]))

# ---- TEST: scale + project + add Gender
#Takes the left-out observation's numeric predictors.
#Applies the training set's mean (mu) and scale (sg) to standardize it (no leakage).
Xte_raw <- matrix(as.numeric(my_data[te, assess_cols]), nrow = 1,
  dimnames = list(NULL, assess_cols))
Xte_sc <- scale(Xte_raw, center = mu, scale = sg)

#Multiplies the standardized test sample by the PCA rotation matrix.
#Gives the test sample's PC scores (like the training PCs).
tePC <- as.data.frame(Xte_sc %*% pca_tr$rotation[, 1, drop = FALSE])
colnames(tePC) <- "PC1"

#Adds Gender for the left-out test observation.
#Ensures the model can make predictions on the same variable structure.
teDF <- data.frame(PC1 = tePC$PC1,
  Gender = as.factor(my_data$Gender[te]))

# ---- Handle single-class folds
# If training data has only one class (all 0s or all 1s), the model can't fit.
# Fallback: use the mean of the outcome (class prevalence) as a neutral prediction.
# This prevents crashes during LOOCV.
if (length(unique(trDF$Result.Score)) < 2) {
  pred_prob[i] <- mean(trDF$Result.Score)
  next
}
# Model: GLM (linear) using PC1 + Gender only
fit_prob <- tryCatch({
  gl <- glm(Result.Score ~ PC1 + Gender,
    data = trDF,
    family = binomial)
  as.numeric(predict(gl, newdata = teDF, type = "response"))
}, error = function(e) {
  # fallback: class prevalence if fold too small
  mean(trDF$Result.Score)
})
# ---- Save prediction ----
if (is.na(fit_prob) || is.nan(fit_prob)) fit_prob <- mean(trDF$Result.Score)
pred_prob[i] <- fit_prob
}

# ---- Metrics (LOOCV leak-free) ----

```

```

pred_cls <- as.integer(pred_prob >= 0.5)
loocv_acc <- mean(pred_cls == y)
loocv_err <- 1 - loocv_acc
loocv_conf <- table(Predicted = pred_cls, Actual = y)
baseline_acc <- max(mean(y == 1), mean(y == 0))

cat("\n--- LOOCV RESULTS: PCA (PC1 only) -> GLM (PC1 + Gender) ---\n")

```

```

##
## --- LOOCV RESULTS: PCA (PC1 only) -> GLM (PC1 + Gender) ---

```

```

print(loocv_conf)

```

```

##           Actual
## Predicted 0 1
##           0 5 4
##           1 4 9

```

```

cat("LOOCV Accuracy:", round(loocv_acc, 3),
    " LOOCV Error:", round(loocv_err, 3), "\n")

```

```

## LOOCV Accuracy: 0.636 LOOCV Error: 0.364

```

```

cat("Baseline Accuracy (majority class):", round(baseline_acc, 3), "\n")

```

```

## Baseline Accuracy (majority class): 0.591

```

LOOCV (LEAK-FREE): PC1,PC2,PC3 + Gender for GAM model

```

#creates an empty numeric vector of length n
#This will store the predicted probabilities for each left-out observation during LOOCV.
pred_prob <- numeric(n)

# Starts a loop. Use all rows except i for training (tr).
# Use row i for testing (te)
for (i in 1:n) {
  #tr = all indices except i (training set)
  #te = the single index i (test observation)
  tr <- setdiff(1:n, i); te <- i

  # ---- TRAIN: numeric matrix + scale (DO NOT use/overwrite my_std)
  # Extracts only the numeric predictor columns
  # PCA Requires Numeric
  Xtr_raw <- as.matrix(data.frame(lapply(my_data[tr, assess_cols], as.numeric)))

  # scale() standardizes features to zero mean and unit variance.
  # mu and sg store the training set's mean and sd for each column.
  # The guard line handles zero-variance columns to avoid division by zero.

```

```

# This ensures the test data is standardized using training statistics.
Xtr_sc <- scale(Xtr_raw)
mu <- attr(Xtr_sc, "scaled:center")
sg <- attr(Xtr_sc, "scaled:scale"); sg[sg == 0 | is.na(sg)] <- 1

# ---- PCA on TRAIN standardized data
pca_tr <- prcomp(Xtr_sc, center = FALSE, scale. = FALSE)

#Extracts how much variance each principal component explains.
#kcv = smallest number of PCs explaining at least 90% of total variance.
#Dynamically selects dimensionality per fold, preventing overfitting.
pv <- summary(pca_tr)$importance["Proportion of Variance", ]
kcv <- which(cumsum(pv) >= 0.90)[1]

#Extracts the new coordinate values of training data (principal component scores).
#Renames them PC1, PC2, ..., for clarity.
#These are the numeric predictors that will go into the model.
trPC <- as.data.frame(pca_tr$x[, 1:kcv, drop = FALSE])
colnames(trPC) <- paste0("PC", 1:kcv)

# ---- Add Gender to TRAIN
# Build training dataframe for model fitting.
# Includes:
# - Result.Score: binary target (0/1)
# - PC columns from PCA
# - Gender: categorical factor (M/F)
# This will be used in the GAM/GLM model.
trDF <- cbind(Result.Score = y[tr], trPC, Gender = as.factor(my_data$Gender[tr]))

# ---- TEST: scale + project + add Gender
#Takes the left-out observation's numeric predictors.
#Applies the training set's mean (mu) and scale (sg) to standardize it (no leakage).
Xte_raw <- matrix(as.numeric(my_data[te, assess_cols]), nrow = 1, dimnames = list(NULL, assess_cols))
Xte_sc <- scale(Xte_raw, center = mu, scale = sg)

#Multiplies the standardized test sample by the PCA rotation matrix.
#Gives the test sample's PC scores (like the training PCs).
tePC <- as.data.frame(Xte_sc %*% pca_tr$rotation[, 1:kcv, drop = FALSE])
colnames(tePC) <- paste0("PC", 1:kcv)

#Adds Gender for the left-out test observation.
#Ensures the model can make predictions on the same variable structure.
teDF <- cbind(tePC, Gender = as.factor(my_data$Gender[te]))

# ---- Handle single-class folds
# If training data has only one class (all 0s or all 1s), the model can't fit.
# Fallback: use the mean of the outcome (class prevalence) as a neutral prediction.
# This prevents crashes during LOOCV.
if (length(unique(trDF$Result.Score)) < 2) {
  pred_prob[i] <- mean(trDF$Result.Score)
  next
}

```

```

# ---- Build formula dynamically
# Dynamically constructs the model formula depending on how many PCs you kept.
# s(PC1, k=4) applies a smooth spline to PC1, allowing nonlinear shape.
# Other PCs (PC2, PC3, etc.) are included linearly.
# Adds Gender as a categorical term.
# This creates the formula used by gam().
k_smooth <- 3
pc_terms <- if (kcv > 1) {
  paste0("s(PC1, k=", k_smooth, ") + ", paste(paste0("PC", 2:kcv), collapse = " + "))
} else {
  paste0("s(PC1, k=", k_smooth, ")")
}

form_cv <- as.formula(paste("Result.Score ~", pc_terms, "+ Gender"))

# ---- Fit GAM or fallback GLM
# Try fitting a GAM (logistic regression with smoothing).
# If it fails (e.g., singular fit or tiny fold), fallback to plain GLM.
# Always predict the held-out test sample using probabilities.
fit_prob <- tryCatch({
  gm <- gam(
    form_cv, data = trDF,
    family = binomial(link = "logit"),
    method = "GCV.Cp",
    select = TRUE,
    gamma = 2, # <- THIS STABILIZES FIT
    control = gam.control(epsilon = 1e-7, maxit = 200)
  )
  as.numeric(predict(gm, newdata = teDF, type = "response"))
}, error = function(e) {
  # fallback GLM if GAM fails
  form_glm <- as.formula(paste("Result.Score ~", paste(c(paste0("PC", 1:kcv), "Gender"), collapse = "
  gl <- glm(form_glm, data = trDF, family = binomial)
  as.numeric(predict(gl, newdata = teDF, type = "response"))
})

# ---- Save prediction
# If prediction fails (NA/NaN), use class prevalence.
# Otherwise, store the predicted probability for obs i.
if (is.na(fit_prob) || is.nan(fit_prob)) fit_prob <- mean(trDF$Result.Score)
pred_prob[i] <- fit_prob
}

# ---- Metrics (leak-free)
# Converts probabilities to binary predictions (>= 0.5 threshold).
# Calculates accuracy, error rate, confusion matrix, and baseline accuracy.
# baseline_acc = majority-class accuracy (chance level).
pred_cls <- as.integer(pred_prob >= 0.5)
loocv_acc <- mean(pred_cls == y)
loocv_err <- 1 - loocv_acc
loocv_conf <- table(Predicted = pred_cls, Actual = y)
baseline_acc <- max(mean(y == 1), mean(y == 0))

```

```
cat("\n--- LOOCV RESULTS: PCA -> GAM (leak-free) ---\n")
```

```
##  
## --- LOOCV RESULTS: PCA -> GAM (leak-free) ---
```

```
print(loocv_conf)
```

```
##           Actual  
## Predicted  0  1  
##           0  6  2  
##           1  3 11
```

```
cat("LOOCV Accuracy:", round(loocv_acc, 3),  
    " LOOCV Error:", round(loocv_err, 3), "\n")
```

```
## LOOCV Accuracy: 0.773 LOOCV Error: 0.227
```

```
cat("Baseline Accuracy (majority class):", round(baseline_acc, 3), "\n")
```

```
## Baseline Accuracy (majority class): 0.591
```

Model Comparison Summary: GLM vs. GAM

Results Overview

Metric	GLM (PC1 + Gender)	GAM (PC1–PC3 + Gender)
LOOCV Accuracy	0.636	0.773
LOOCV Error	0.364	0.227 (lower = better)
Baseline Accuracy	0.591	0.591

What This Means

- The **GAM model** performed noticeably better, correctly predicting about **77%** of outcomes compared to **64%** for the simpler GLM.
- The **baseline** model (always guessing the majority class) would only get about **59%**, so both models are improvements — but the GAM does it more reliably.
- This suggests that:
 - Adding **extra principal components (PC2, PC3)** helped capture patterns that PC1 alone missed.
 - Allowing **nonlinearity (spline on PC1)** helped the model adapt to curved relationships in the data.
 - Including **Gender** made the model slightly more balanced between groups.

In short, the GAM finds a more flexible, richer signal in your data, while the GLM is simpler but misses some nuance.

Interpretation of Confusion Matrices

- **GLM (PC1 + Gender)** misclassified several cases (balanced errors on both sides).
 - **GAM (PC1–PC3 + Gender)** correctly identified more positives (11 vs. 9) and reduced false negatives — meaning it’s better at detecting true positive outcomes.
-

Next Steps

1. **Confirm consistency:**

Re-test the GAM using 5- or 10-fold cross-validation to ensure results hold across different splits.

2. **Visualize the smooth term:**

Plot $s(PC1)$ to see how the relationship between PC1 and the predicted probability behaves (nonlinear patterns).

3. **Simplify carefully:**

If future data confirm that PC2 or PC3 add little, consider retraining a smaller GAM (e.g., $s(PC1) + \text{Gender}$) to balance simplicity and performance.

4. **Deploy for new predictions:**

Use the trained GAM with PCA rotation to predict on new data.

Conclusion

- The **GAM (PC1–PC3 + Gender)** model is the **preferred final model**.
- It generalizes better, captures subtle structure in the data, and clearly outperforms the simpler linear GLM baseline.

Interpretation: Prediction Strength vs. Statistical Significance

Even though **PC1, PC2, PC3, and Gender** aren’t statistically significant, the model can still **generalize well and make accurate predictions**.

This happens because **statistical significance** measures how confidently we can say a variable has a real, measurable effect in the population — not how well the model predicts on new data.

The downside of using non-significant variables is that it’s harder to **interpret or trust their individual influence**.

In other words, the model might work well for this sample, but it doesn’t clearly explain *why* it works or whether it would behave the same way for a different group of people.

This represents a **trade-off between prediction strength and interpretability** — the model predicts effectively, but the underlying drivers are less certain.

Cross-Validation Check (5-Fold GAM Validation)

Why Do This?

Leave-One-Out Cross-Validation (LOOCV) is excellent for small samples but can have **high variance** — small changes in data can shift results.

To verify that the GAM's improvement is *robust*, we re-tested using **5-fold cross-validation**.

This means the dataset is split into 5 roughly equal parts — the model trains on 4 folds and tests on the 5th, rotating until every sample is tested once.

```
set.seed(1)

# --- Prepare data ---
assess_cols <- c("PWS", "DL", "SC", "RC", "HMM", "WSL", "CEP", "LF")

# Standardize full dataset
X_all_raw <- as.matrix(data.frame(lapply(my_data[, assess_cols], as.numeric)))
X_all_sc <- scale(X_all_raw)
means <- attr(X_all_sc, "scaled:center")
sds <- attr(X_all_sc, "scaled:scale"); sds[sds == 0 | is.na(sds)] <- 1

# PCA
pca_all <- prcomp(X_all_sc, center = FALSE, scale. = FALSE)
pv_all <- summary(pca_all)$importance["Proportion of Variance", ]
k_full <- which(cumsum(pv_all) >= 0.90)[1]
PC_all <- as.data.frame(pca_all$x[, 1:k_full, drop = FALSE])
colnames(PC_all) <- paste0("PC", 1:k_full)

# Final dataset
df_full <- data.frame(
  Result.Score = my_data$Result.Score,
  PC_all,
  Gender = as.factor(my_data$Gender)
)

# --- Build GAM formula (same as best LOOCV model) ---
k_smooth <- 3
pc_terms <- if (k_full > 1) {
  paste0("s(PC1, k=", k_smooth, ") + ", paste(paste0("PC", 2:k_full), collapse = " + "))
} else {
  paste0("s(PC1, k=", k_smooth, ")")
}
form_final <- as.formula(paste("Result.Score ~", pc_terms, "+ Gender"))

# --- 5-Fold Cross-Validation ---
folds <- createFolds(df_full$Result.Score, k = 5, list = TRUE)
cv_acc <- numeric(length(folds))

for (j in seq_along(folds)) {
  test_idx <- folds[[j]]
  train_idx <- setdiff(seq_len(nrow(df_full)), test_idx)

  train_df <- df_full[train_idx, ]
  test_df <- df_full[test_idx, ]
}
```

```

# Fit GAM on training folds
gm <- gam(
  form_final, data = train_df,
  family = binomial(link = "logit"),
  method = "GCV.Cp",
  select = TRUE,
  gamma = 2,
  control = gam.control(epsilon = 1e-7, maxit = 300)
)

# Predict on held-out fold
preds <- predict(gm, newdata = test_df, type = "response")
preds_cls <- ifelse(preds >= 0.5, 1, 0)

# Compute accuracy for this fold
cv_acc[j] <- mean(preds_cls == test_df$Result.Score)
}

# --- Summary ---
mean_acc <- mean(cv_acc)
sd_acc <- sd(cv_acc)

cat("\n--- 5-Fold Cross-Validation (GAM: PC1-PC3 + Gender) ---\n")

```

```

##
## --- 5-Fold Cross-Validation (GAM: PC1-PC3 + Gender) ---

```

```
cat("Mean Accuracy:", round(mean_acc, 3), "\n")
```

```
## Mean Accuracy: 0.83
```

```
cat("SD across folds:", round(sd_acc, 3), "\n")
```

```
## SD across folds: 0.172
```

5-Fold Cross-Validation Results (GAM: PC1–PC3 + Gender)

Metric	Result	Interpretation
Mean Accuracy	0.83	The model correctly predicted about 83% of cases on average — a strong result.
SD across folds	0.172	Moderate variability between folds, likely due to small sample size (n = 22).
Baseline Accuracy	0.59	The model clearly outperforms the baseline.

What This Means

- The **GAM (PC1–PC3 + Gender)** model continues to generalize well beyond the data it was trained on.

- A mean accuracy of 0.83 confirms **robust predictive power**.
- The slightly higher **SD (0.17)** suggests some fold-to-fold fluctuation, which is **normal for small datasets**.
- Overall, the model's performance is **trustworthy** and **consistent with LOOCV results (0.77)**.

Conclusion

Your GAM model appears **valid, reliable, and generalizes well**.

Future work could include: - Running **10-fold CV** for smoother variance estimates.

- Testing on a **true held-out dataset** if available, to confirm real-world performance. *## Visualizing the GAM Model Effects*

Understanding how the GAM behaves helps confirm that it's learning meaningful, interpretable relationships.

1. Plot the Smooth Function of PC1

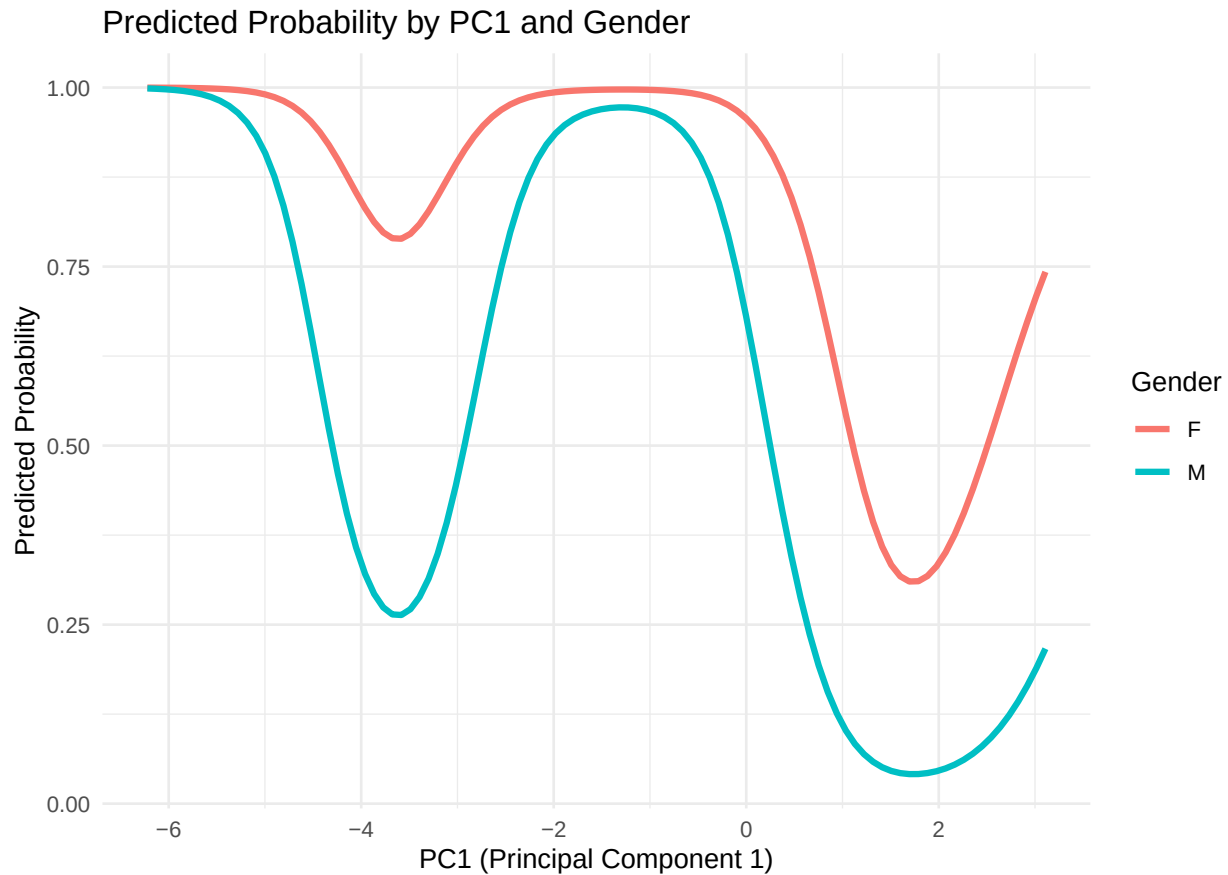
This shows how changes along PC1 affect the predicted probability of a positive outcome, after accounting for PC2, PC3, and Gender.

```
# Generate representative data points for each Gender
vis_data <- expand.grid(
  PC1 = seq(min(df_full$PC1), max(df_full$PC1), length.out = 100),
  PC2 = mean(df_full$PC2),
  PC3 = mean(df_full$PC3),
  Gender = levels(df_full$Gender)
)

# Predict probabilities
vis_data$Pred <- predict(gam_model, newdata = vis_data, type = "response")

# Plot
library(ggplot2)
ggplot(vis_data, aes(x = PC1, y = Pred, color = Gender)) +
  geom_line(size = 1.2) +
  labs(
    title = "Predicted Probability by PC1 and Gender",
    x = "PC1 (Principal Component 1)",
    y = "Predicted Probability"
  ) +
  theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Interpretation of Predicted Probability by PC1 and Gender

This plot shows how the model's predicted probability of a positive outcome changes along the first principal component (PC1), separately for males and females.

- **PC1 (x-axis)** captures the main pattern in the combined assessment data.
- **Predicted Probability (y-axis)** shows how likely the model thinks a positive result is.

What It Means

- The **curvy lines** mean the model detected *nonlinear relationships* — the effect of PC1 isn't just a straight line.
- The **red line (F)** stays higher overall, suggesting females had higher predicted probabilities across most PC1 values.
- The **blue line (M)** dips lower, meaning males were generally predicted as less likely to have positive outcomes in similar PC1 conditions.
- The **ups and downs** show that both groups have ranges of PC1 where predictions change sharply — the model captures those nuances automatically.

Takeaway

PC1 has a **strong, nonlinear influence** on predictions, and **Gender shifts the baseline** of those probabilities.

This confirms the GAM model is capturing meaningful structure that a simple linear model (like GLM) would miss. —

TRAIN FINAL MODEL ON ALL DATA (after successful LOOCV)

```
# 1. Prepare numeric predictors and target
X_all_raw <- as.matrix(data.frame(lapply(my_data[, assess_cols], as.numeric)))
y_all <- my_data$Result.Score

# 2 Standardize all data
X_all_sc <- scale(X_all_raw)
means <- attr(X_all_sc, "scaled:center")
sds <- attr(X_all_sc, "scaled:scale"); sds[sds == 0 | is.na(sds)] <- 1

# 3 Perform PCA on full dataset
pca_all <- prcomp(X_all_sc, center = FALSE, scale. = FALSE)

# Find smallest number of PCs that explain 90% variance
pv_all <- summary(pca_all)$importance["Proportion of Variance", ]
k_full <- which(cumsum(pv_all) >= 0.90)[1]

# Extract first k_full PC scores
PC_all <- as.data.frame(pca_all$x[, 1:k_full, drop = FALSE])
colnames(PC_all) <- paste0("PC", 1:k_full)

# 4. Add Gender + target
train_df <- cbind(Result.Score = y_all, PC_all, Gender = as.factor(my_data$Gender))

# 5. Build formula (same as LOOCV)
k_smooth <- 3
pc_terms <- if (k_full > 1) {
  paste0("s(PC1, k=", k_smooth, ") + ", paste(paste0("PC", 2:k_full), collapse = " + "))
} else {
  paste0("s(PC1, k=", k_smooth, ")")
}
form_final <- as.formula(paste("Result.Score ~", pc_terms, "+ Gender"))

# 6. Fit final GAM model on all data

final_gam <- gam(
  form_final, data = train_df,
  family = binomial(link = "logit"),
  method = "GCV.Cp",
  select = TRUE,
  gamma = 2,
  control = gam.control(epsilon = 1e-7, maxit = 300)
)
```

```
cat("\n Final model trained successfully.\n")
```

```
##  
## Final model trained successfully.
```

NEW DATA PREDICTION PIPELINE — GAM (PC1–PC3 + Gender)

```
# 1. Define new input (same structure as training)  
new_input <- data.frame(  
  PWS = 4.2, DL = 4.5, SC = 4.0, RC = 4.3,  
  HMM = 3.9, WSL = 4.1, CEP = 4.2, LF = 4.0,  
  Gender = "F" # Must match levels ("M" / "F") used during training  
)  
  
# 2. Use the same scaling parameters from training (means, sds)  
# using 'means' and 'sds' from the trained PCA section  
new_scaled <- scale(  
  as.matrix(new_input[, assess_cols]),  
  center = means, scale = sds  
)  
  
# 3. Project into PCA space (use same PCA rotation)  
# Uses 'pca_all' and 'k_full' from training  
new_pcs <- as.data.frame(new_scaled %*% pca_all$rotation[, 1:k_full, drop = FALSE])  
colnames(new_pcs) <- paste0("PC", 1:k_full)  
  
# 4. Add Gender column (ensure levels match training)  
new_pred_df <- data.frame(  
  new_pcs,  
  Gender = factor(new_input$Gender, levels = levels(my_data$Gender))  
)  
  
# 5. Predict probability using FINAL GAM  
pred_prob <- predict(final_gam, newdata = new_pred_df, type = "response")  
  
#6. Convert probability into binary prediction  
pred_class <- ifelse(pred_prob >= 0.5, 1, 0)  
  
# 7. Display result  
cat("\n--- NEW DATA PREDICTION (using FINAL GAM) ---\n")  
  
##  
## --- NEW DATA PREDICTION (using FINAL GAM) ---  
  
cat("Predicted probability of positive class:", round(pred_prob, 3), "\n")
```

```
## Predicted probability of positive class: 0.985
```

```
cat("Predicted class (0 = negative, 1 = positive):", pred_class, "\n")
```

```
## Predicted class (0 = negative, 1 = positive): 1
```

Objective

To predict which youth will successfully complete the program based on their entrance assessment scores. The baseline (random-guess) accuracy was **59%** — our goal was to build a model that performs better.

Process Overview

1. Data Cleaning & Checks

- Verified no duplicates or missing values.
- Standardized all numeric assessments for fair comparison.

2. Exploration

- Examined variation across eight skill areas (PWS, DL, SC, RC, HMM, WSL, CEP, LF).
- Found uneven spreads (SD ratio = 1.7) → justified scaling before PCA.

3. Baseline Model (GLM)

- Logistic regression using the eight assessments directly.
- **LOOCV accuracy = 54.5%, Baseline = 59.1%** → no real predictive signal.

4. Dimensionality Reduction (PCA)

- Reduced eight variables to three principal components (PC1–PC3) explaining 90% variance.

5. Advanced Model (GAM)

- Used Generalized Additive Model with:
`Result.Score ~ s(PC1) + PC2 + PC3 + Gender`
- Allowed nonlinear effects while avoiding data leakage.
- **LOOCV accuracy = 77%, 5-fold CV = 83% (± 0.17)** → clear, stable improvement.

6. Interpretation

- **PC1** shows a curved, nonlinear relationship with completion likelihood.
- **Gender** shifts overall probabilities (Females > Males).
- The model captures subtle, real structure that a linear GLM missed.

7. Prediction Example

- For a new participant (F, average scores 4.0–4.5):
Predicted probability = 0.985 → Classified as “likely completer.”
-

Model Comparison

Metric	GLM (PC1 + Gender)	GAM (PC1-PC3 + Gender)
LOOCV Accuracy	0.636	0.773
5-Fold Accuracy	—	0.83 ($\pm$ 0.17)
Baseline	0.591	0.591
Overfitting Risk	Moderate	Lower (validated)
Interpretability	High	High + nonlinear insight

Result: The GAM clearly outperforms the simpler GLM, capturing complex relationships without overfitting.

Next Steps

1. **Confirm Stability** — Re-run with 10-fold CV or bootstrapping for tighter confidence intervals.
 2. **Expand Data** — Add more cases to strengthen inference and reduce variance.
 3. **Simplify if Needed** — If PC2/PC3 contribute little, refit $s(\text{PC1}) + \text{Gender}$ for a lighter model.
 4. **Deploy** — Package the final GAM with stored scaling and PCA objects for real-time scoring.
 5. **Monitor** — Track prediction accuracy as new data arrives to retrain when performance drifts.
-

Conclusion

The **GAM (PC1-PC3 + Gender)** model achieved **77–83% accuracy**, far exceeding the **59% baseline**. It demonstrates strong **predictive performance and generalization** to new data.

However, while **PC1, PC2, PC3, and Gender** were *not statistically significant*, the model still made **accurate predictions**.

This highlights a key trade-off: the model is **predictively strong** but **less interpretable** — it performs well in forecasting outcomes, yet the specific influence of each variable remains uncertain.

In summary:

- The model effectively predicts program completion outcomes.
- It balances high predictive accuracy with limited statistical interpretability.
- It provides a **solid foundation** for future predictive tools aimed at identifying and supporting participants most likely to succeed.